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COUPLING OF OPTIMIZATION AND INVERSE PROBLEM FOR AERODYNAMIC SHAPE DESIGN

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Abstract

A classical inverse problem in fluid dynamics, i.e., determine the geometry of an airfoil given the aerodynamic load, is solved coupled to an adjoint optimization method. By the adjoint technique we determine the prescribed aerodynamic load so to maximize a given functional. As an example we design the loading of turbomachine bladings for maximum thrust, based on a simplified compressible flow model.

Prologue

Usual optimization methods iterate on the geometry of a certain flow problem to determine a configuration minimizing or maximizing a given objective function, i.e., for each optimization step a direct problem is solved[1]. In the present study we propose to iterate on inverse problems[2], motivated by the fact that in such formulation fluid-dynamic constraints on the solution are very easily imposed.

As a practical example, we investigate the application of an adjoint optimization method to the flow design of turbomachinery bladings, based on a simplified flow model. The blades of the turbomachinery are modeled as flow surfaces of zero thickness which exert forces on the fluid flow. This approximation introduces volume forces in the compressible Euler equations, which is the model adopted for the flow.

Our method is such that instead of modifying the shape of the flow surfaces modeling the blades, we give the force which the blades exert on the flow, and let the geometry accommodate such distribution of forces. Then the volume force distribution itself varies based on the functional gradient, so that for example thrust is maximized.

Mathematically our problem is formulated as a distributed parameter control where the maximum conditions based on the adjoint formulation are written for the inverse problem.

In this paper we include the problem formulation as well as preliminary applications to turbomachines row design.

Flow model

The flow deflection through rotors and stators of a turbomachine is the result of the forces that rotors and stators blades exert on the flow. An axis-symmetric model of a turbomachine can be set up by replacing the blade rows with volume forces. We assume that the blade rows have vanishing thickness and infinite solidity, so that the single blade coincides with a stream surface. Thus, in the case of an inviscid flow, the effect of solid blades is modeled by volume forces orthogonal to stream surfaces.

Let

$$\mathbf{F} = F^x i + F^r \xi + F^\vartheta \eta \tag{1}$$

be the volume force, where i, ξ and η are the unit vectors pertinent to the axial, radial and tangential directions in a cylindrical frame of reference $(xi,\ r\xi,\ \vartheta\eta)$.

The distribution of the tangential component $F^\vartheta = F^\vartheta(x, r)$ is the function to be optimized, the same way the shape of a wall is typically optimized in usual optimization algorithms.

The geometry of the blades, represented by 2D manifolds

$$\Theta(x, r, \vartheta) = 0 \tag{2}$$

is found by solving

$$(\mathbf{q} - j\omega r\eta) \cdot \nabla\Theta = 0 \tag{3}$$

as the blades are to be stream surfaces of the absolute or relative motion for stators and rotors respectively. In the equation above $\mathbf{q} = u\,i + w\,\xi + v\,\eta$ is the flow velocity vector, ω is the angular velocity of rotors and $j = 0$ for stators, $j = 1$ for rotors.

The components of the volume force F^x and F^r are determined enforcing the blade manifolds to be orthogonal to the volume forces

$$\mathbf{F} \times \nabla\Theta = 0. \tag{4}$$

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which implies

$$F^x = r\frac{\Theta_x}{\Theta_\vartheta}F^\vartheta \tag{5}$$

and

$$F^r = r\frac{\Theta_r}{\Theta_\vartheta}F^\vartheta \tag{6}$$

Flow-field solution

In this section we detail the solution technique of the inverse problem taking as known $F^\vartheta(x,r)$. However, it should be noted that this distribution is updated during the optimization in order to maximize the cost function, which in our case is thrust.

The solution of the inverse design problem is obtained by means of a time dependent process. The blades can be seen as deformable and impermeable surfaces constrained to the leading edge. They move like fastened sails waving under the wind effect. An initial configuration of such surface is guessed. The following transient is described by integrating in time the equations governing the time dependent flow motion. At the end of the transient the blades assume the shape that solve the inverse problem.

In a cylindrical frame of reference, the compressible Euler equations with volume forces acting on the fluid are

$$\frac{\partial \mathbf{U}}{\partial t} + \frac{\partial \mathbf{A}}{\partial x} + \frac{\partial \mathbf{B}}{\partial r} + \mathbf{Q} = 0 \tag{7}$$

where

$$\mathbf{U} = \begin{Bmatrix} \rho \\ \rho u \\ \rho v \\ \rho w \\ e \end{Bmatrix} \qquad \mathbf{A} = \begin{Bmatrix} \rho u \\ p + \rho u^2 \\ \rho uv \\ \rho uw \\ u(p+e) \end{Bmatrix}$$

$$\mathbf{B} = \begin{Bmatrix} \rho w \\ \rho uw \\ \rho vw \\ p + \rho w^2 \\ w(p+e) \end{Bmatrix}$$

$$\mathbf{Q} = \begin{Bmatrix} \dfrac{\rho w}{r} + \rho u\alpha \\ \dfrac{\rho uw}{r} - F^x + \rho u^2\alpha \\ 2\dfrac{\rho vw}{r} - F^\theta \\ \dfrac{\rho(v^2 - w^2)}{r} - F^r \\ \dfrac{w(p+e)}{r} - \mathbf{F}\cdot\mathbf{q} + u(p+e)\alpha \end{Bmatrix}$$

and as usual ρ is density, p pressure, e total internal energy per unit volume.

The boundary conditions at the entry section are the two ratios between the velocity components, the total pressure and the total temperature when the flow is subsonic, while all the flow properties are prescribed if the flow is supersonic; at the exit section the static pressure is prescribed if the flow is subsonic, while no boundary conditions are needed when the flow is supersonic.

The blades blockage is taken into account by the terms containing the coefficient α, with

$$\alpha = \frac{\partial(\log \Xi)}{\partial x}$$

and with Ξ being the free passage per unit radius:

$$\Xi = 2\pi r - T$$

$T = T(x,r)$ is the sum of the estimated blades thickness, including the boundary layers.

The system of eq. 7 is integrated in time using a finite volume formulation based on an approximate Riemann solver[3] to compute the fluxes at cell interfaces. Second order spatial accuracy is obtained by an ENO class method[4].

A blade surface changes its shape during the transient to obey the impermeable wall condition. Let us express eq.2 as

$$\Theta(x,r,\vartheta,t) = \vartheta - g(x,r,t) = 0 \tag{8}$$

so that eqs. 5-6 become

$$F^x = -rg_xF^\vartheta \tag{9}$$

$$F^r = -rg_rF^\vartheta. \tag{10}$$

Flow particles on $\Theta(x,r,\vartheta,t) = 0$ must remain on the manifold for the impermeable wall condition. It follows that during the transient the Langragian derivative of the function $\Theta(x,r,\vartheta,t)$ has to be null

$$\frac{d\Theta}{dt} = \Theta_t + (q - j\omega r\eta)\cdot\nabla\Theta = 0 \tag{11}$$

that can be written as

$$g_t = -ug_x - wg_r + \frac{v - j\omega r}{r} \tag{12}$$

with $j = 0$ for stators and $j = 1$ for rotors. The above equation is solved coupled to the Euler equations, and it is integrated in time upwinding the spatial derivatives of g according to u and w.

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Flow equations adjoint

In this section we derive the functional differential using the adjoint technique. As functional we consider the conventional thrust expressed as

$$T(F^\vartheta) = \left[\int_{r_h}^{r_t} (p+\rho u^2) r\, dr\right]_{out} +$$

$$- \left[\int_{r_h}^{r_t} (p+\rho u^2) r\, dr\right]_{in} = \int_{\Gamma_{io}} H(U)\, d\Gamma \quad (13)$$

where F^ϑ, the distribution of tangential forces, is the control, r_t and r_h are the tip and hub radius respectively. The maximum of T is constrained by the the steady state Euler equations

$$\mathbf{E}(F^\vartheta) = \mathbf{A}_x + \mathbf{B}_r + \mathbf{Q} = 0 \quad (14)$$

and by the kinematic constraint on the blades

$$\mathbf{G}(\mathbf{U}(F^\vartheta)) = u g_x + w g_r - \frac{v - j\omega r}{r} = 0 \quad (15)$$

In order to solve such constrained maximization problem we introduce the Lagrangian function

$$\mathcal{L}(\mathbf{U}, g, F^\vartheta, \Lambda, \mu) = \int_{\Gamma_{io}} \mathbf{H}(\mathbf{U})\, d\Gamma +$$

$$+ \int_\Omega {}^t\Lambda\, \mathbf{E}(\mathbf{U}, F^\vartheta, g)\, d\Omega + \int_\Omega \mu\, G(\mathbf{U}, g)\, d\Omega \quad (16)$$

where ${}^t\Lambda(x,r) = \{\lambda_1, \lambda_2, \lambda_3, \lambda_4, \lambda_5\}$ and $\mu = \mu(x,r)$ are Lagrange multipliers. The Lagrangian will allow us to treat the maximization problem as an unconstrained problem, so that a stationary point is found when the variation of $\mathcal{L}$ with respect to all its arguments, that are now considered independent functions, is 0. Let us compute $\delta\mathcal{L}$. We have

$$\delta\mathcal{L} = \delta\mathcal{L}_U + \delta\mathcal{L}_{F^\vartheta} + \delta\mathcal{L}_g + \delta\mathcal{L}_\Lambda + \delta\mathcal{L}_\mu \quad (17)$$

with

$$\delta\mathcal{L}_U = \int_{\Gamma_{io}} \frac{\partial \mathbf{H}}{\partial \mathbf{U}} \delta\mathbf{U}\, d\Gamma + \int_\Omega {}^t\Lambda\, \delta\mathbf{E}_U\, d\Omega +$$

$$+ \int_\Omega \mu \frac{\partial G}{\partial \mathbf{U}} \delta\mathbf{U}\, d\Omega \quad (18)$$

$$\delta\mathcal{L}_\Lambda = \int_\Omega {}^t\delta\Lambda\, \mathbf{E}(\mathbf{U}, F^\vartheta, g)\, d\Omega \quad (19)$$

$$\delta\mathcal{L}_\mu = \int_\Omega G(\mathbf{U}, g)\delta\mu\, d\Omega \quad (20)$$

$$\delta\mathcal{L}_{F^\vartheta} = \int_\Omega {}^t\Lambda \frac{\partial \mathbf{Q}}{\partial F^\vartheta} \delta F^\vartheta\, d\Omega \quad (21)$$

$$\delta\mathcal{L}_g = \int_\Omega {}^t\Lambda\, \delta\mathbf{Q}_g\, d\Omega + \int_\Omega \mu\, \delta G_g\, d\Omega \quad (22)$$

The vectors $\frac{\partial \mathbf{Q}}{\partial F^\vartheta}$ and $\frac{\partial G}{\partial \mathbf{U}}$ are easily computed; $\frac{\partial \mathbf{H}}{\partial \mathbf{U}}$ is the derivative of the difference of the fluxes component in the direction x, taken at inlet and outlet sections.

In order to have $\delta\mathcal{L} = 0$, all the single contributions to $\delta\mathcal{L}$ must be 0 at the maximum, so that to find a stationary point we enforce

$$\delta\mathcal{L}_U = 0 \qquad \delta\mathcal{L}_\Lambda = 0 \qquad \delta\mathcal{L}_\mu = 0 \qquad \delta\mathcal{L}_g = 0$$

In general this results in $\delta\mathcal{L}_{F^\vartheta} \neq 0$. To reach the maximum we take δF^ϑ such that $\delta\mathcal{L} = \delta\mathcal{L}_{F^\vartheta} > 0$, for example using a conjugate gradient method, as explained in the following.

We can manipulate the second integral in eq. 16 obtaining

$$\delta\mathcal{L}_U =$$

$$= \int_{\Gamma_{io}} \frac{\partial \mathbf{H}}{\partial \mathbf{U}} \delta\mathbf{U}\, d\Gamma + \int_\Sigma {}^t\Lambda\, (\mathbf{A}_U\, n_x + \mathbf{B}_U\, n_r) \delta\mathbf{U}\, d\sigma +$$

$$- \int_\Omega ({}^t\Lambda_x\, \mathbf{A}_U + {}^t\Lambda_r \mathbf{B}_U) \delta\mathbf{U}\, d\Omega + \int_\Omega {}^t\Lambda \frac{\partial \mathbf{Q}}{\partial \mathbf{U}} \delta\mathbf{U}\, d\Omega +$$

$$+ \int_\Omega \mu \frac{\partial G}{\partial \mathbf{U}} \delta\mathbf{U}\, d\Omega \quad (23)$$

where Σ is the entire border of the flow field Ω, and $\mathbf{A}_U$, $\mathbf{B}_U$ and $\mathbf{Q}_U$ are Jacobian matrices not given for sake of conciseness.

From the condition $\delta\mathcal{L}_U = 0$ we obtain the so called "adjoint" of the Euler equations and its boundary conditions, that is

$${}^t\Lambda_x\, \mathbf{A}_U + {}^t\Lambda_r \mathbf{B}_U - {}^t\Lambda \frac{\partial \mathbf{Q}}{\partial \mathbf{U}} - \mu \frac{\partial G}{\partial \mathbf{U}} = 0 \text{ in } \Omega \quad (24)$$

and

$$\left[\frac{\partial \mathbf{H}^*}{\partial \mathbf{U}} + {}^t\Lambda\, (\mathbf{A}_U\, n_x + \mathbf{B}_U\, n_r)\right] \delta\mathbf{U} = 0 \text{ on } \Sigma \quad (25)$$

where $H^* = H$ for the inlet and the outlet, and $H^* = 0$ elsewhere. From the above equation we derive the boundary condition for the adjoint, as explained in the following.

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The condition $\delta\mathcal{L}_g = 0$ yields

$$\delta\mathcal{L}_g = \int_{\Omega_b} \mu\,\delta G_g\,d\Omega + \int_{\Omega_b} {}^t\Lambda\,\delta\mathbf{Q}_g\,d\Omega =$$

$$\int_{\Sigma_b} \mu\,(\mathbf{q}\cdot\mathbf{n})\delta g\,d\sigma - \int_{\Omega_b} [(\mu u)_x + (\mu w)_r]\,\delta g\,d\Omega +$$

$$+ \int_{\Omega_b} {}^t\Lambda\,\delta\mathbf{Q}_g\,d\Omega = 0 \tag{26}$$

where it should be noted that the domain of integration Ω_b and the bounding curve Γ_b are those relative to the blades. The last integral in the above equation is

$$\int_{\Omega_b} {}^t\Lambda\,\delta\mathbf{Q}_g\,d\Omega = \int_{\Omega_b} {}^t\Lambda\,\mathbf{K}\nabla(\delta g)\,d\Omega =$$

$$= \int_{\Sigma_b} ({}^t\Lambda\,\mathbf{K})\cdot\mathbf{n}\,\delta g\,d\sigma - \int_{\Omega_b} \nabla\cdot({}^t\Lambda\,\mathbf{K})\delta g\,d\Omega \tag{27}$$

where $\mathbf{n} = (n_x, n_y)$ and

$$\mathbf{K} = rF^\vartheta \begin{Bmatrix} 0 & 0 \\ 1 & 0 \\ 0 & 0 \\ 0 & 1 \\ u & w \end{Bmatrix}$$

Hence, the adjoint of the kinematic constraint is

$$(\mu u)_x + (\mu w)_r - \nabla\cdot({}^t\Lambda\,\mathbf{K}) = 0 \text{ in } \Omega_b \tag{28}$$

and

$$[\mu\,(\mathbf{q}\cdot\mathbf{n}) + ({}^t\Lambda\,\mathbf{K})\cdot\mathbf{n}]\,\delta g = 0 \text{ on } \Gamma_b \tag{29}$$

which yields the boundary conditions for eq.28 as explained in the following. The adjoint equation of the kinematic constraint is coupled to eq.24 the same way the kinematic constraint is coupled to the flow equations.

Note that the variations of $\mathcal{L}$ with respect to the Lagrange multipliers Λ and μ simply yield the flow equations and the kinematic constraint respectively.

Finally, we are left with

$$\delta\mathcal{L} = \delta\mathcal{L}_{F^\vartheta} = \int_{\Omega_b} {}^t\Lambda\,\frac{\partial\mathbf{Q}}{\partial F^\vartheta}\,\delta F^\vartheta\,d\Omega \tag{30}$$

This functional depends on $\mathbf{U}$, Λ, μ; variables that satisfy the flow equations, the kinematic constraint and the respective adjoints. Therefore if we update the present distribution of F^ϑ with

$$\delta F^\vartheta = \varrho\,{}^t\Lambda\,\frac{\partial\mathbf{Q}}{\partial F^\vartheta}$$

taking $\varrho > 0$, then $\delta\mathcal{L} > 0$. Iterating such procedure the maximum is eventually reached.

This method, namely the gradient method, has a very slow convergence rate. Better convergence rates are obtained with the conjugate gradient method[5], in which the correction to F^ϑ at the iterate k is

$$(\delta F^\vartheta)^k = \varrho\left[({}^t\Lambda\,\frac{\partial\mathbf{Q}}{\partial F^\vartheta})^k - \beta^{k-1}(\delta F^\vartheta)^{k-1}\right]$$

with

$$\beta^{k-1} =$$

$$\frac{\int_{\Omega_b}[({}^t\Lambda\,\frac{\partial\mathbf{Q}}{\partial F^\vartheta})^k - ({}^t\Lambda\,\frac{\partial\mathbf{Q}}{\partial F^\vartheta})^{k-1}]\,({}^t\Lambda\,\frac{\partial\mathbf{Q}}{\partial F^\vartheta})^k d\Omega}{\int_{\Omega_b}[({}^t\Lambda\,\frac{\partial\mathbf{Q}}{\partial F^\vartheta})^{k-1}]^2\,d\Omega}$$

Flow adjoint boundary conditions

We detail here the method to derive the boundary conditions for the flow adjoint equation and the kinematic constraint adjoint equation.

Let $\mathbf{n} = (0,-1,0)$ at the inlet, so that eq.25 reduces to

$$\frac{\partial\mathbf{H}^*}{\partial\mathbf{U}}\delta\mathbf{U} - {}^t\Lambda\,\frac{\partial\mathbf{A}}{\partial\mathbf{U}}\delta\mathbf{U} = 0 \tag{31}$$

As the flow variables $\mathbf{U}$ are to respect given conditions at the boundaries, the variation $\delta\mathbf{U}$ at the inlet is such that the boundary conditions on $\mathbf{U}$ are still satisfied. For example, if the inlet flow is supersonic all of the components of $\mathbf{U}$ are given. In this case $\delta\mathbf{U} = 0$ and consequently there is no boundary condition on Λ.

In the case of subsonic inlet four boundary conditions must be provided, for example

$$dS = 0, \qquad dT^o = 0, \qquad d\sigma = 0, \qquad d\epsilon = 0 \tag{32}$$

where

$$S = \log\frac{p}{\rho} - 2\kappa\log\rho \tag{33}$$

$$T^o = \frac{p}{\rho} + \frac{\kappa}{\gamma}(u^2+v^2+w^2) = -\frac{2\kappa^2}{\gamma}V^2 + 2\kappa\frac{u_5}{u_1} \tag{34}$$

$$\sigma = \frac{v}{u} = \frac{u_3}{u_2}, \qquad \epsilon = \frac{w}{u} = \frac{u_4}{u_2} \tag{35}$$

where we set $\mathbf{U} = (\rho, \rho u, \rho v, \rho w, e) = (u_1, u_2, u_3, u_4, u_5)$, $\kappa = \dfrac{\gamma-1}{2}$ and γ is the specific

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heats ratio. We have

$$\left\{\begin{array}{l} \delta S = \dfrac{\partial S}{\partial u_1}\delta u_1 + \dfrac{\partial S}{\partial u_2}\delta u_2 + \dfrac{\partial S}{\partial u_3}\delta u_3 + \\ + \dfrac{\partial S}{\partial u_4}\delta u_4 + \dfrac{\partial S}{\partial u_5}\delta u_5 = 0 \\ \delta T = \dfrac{\partial T}{\partial u_1}\delta u_1 + \dfrac{\partial T}{\partial u_2}\delta u_2 + \dfrac{\partial T}{\partial u_3}\delta u_3 + \\ + \dfrac{\partial T}{\partial u_4}\delta u_4 + \dfrac{\partial T}{\partial u_5}\delta u_5 = 0 \\ \delta\sigma = \dfrac{\partial \sigma}{\partial u_2}\delta u_2 + \dfrac{\partial \sigma}{\partial u_3}\delta u_3 = 0 \\ \delta\epsilon = \dfrac{\partial \epsilon}{\partial u_2}\delta u_2 + \dfrac{\partial \epsilon}{\partial u_4}\delta u_4 = 0 \end{array}\right. \tag{36}$$

Selecting δu_2 as the independent variation, from eq.36 we obtain

$$\delta\mathbf{U} = \left\{\begin{array}{c} \dfrac{u_1 V^2}{u_2\left((1+\gamma\kappa)V^2 - 2\gamma\kappa u_1 u_5\right)} \\ 1 \\ \dfrac{u_3}{u_2} \\ \dfrac{u_4}{u_2} \\ \dfrac{V^2\left((1-4\kappa^2)u_1 u_5 + 2\kappa^2 V^2\right)}{u_1 u_2\left((1+\gamma\kappa)V^2 - 2\gamma\kappa u_1 u_5\right)} \end{array}\right\}\delta u_2 = \mathbf{J}_i\,\delta u_2 \tag{37}$$

and from eq.31 we have

$$\left[\frac{\partial \mathbf{H}^*}{\partial \mathbf{U}} - {}^t\Lambda\,\frac{\partial \mathbf{A}}{\partial \mathbf{U}}\right]\mathbf{J}_i\,\delta u_2 = 0 \tag{38}$$

so that for a generic increment it must be

$$\left[\frac{\partial \mathbf{H}^*}{\partial \mathbf{U}} - {}^t\Lambda\,\frac{\partial \mathbf{A}}{\partial \mathbf{U}}\right]\mathbf{J}_i = 0$$

which is a scalar relation to be satisfied by the components of Λ. Note that at inlet we have four conditions for the flow problem, and one for the adjoint equations.

At the outlet the situation is specular. For supersonic outlet, we pose $\Lambda = 0$, whereas for subsonic outlet with given static pressure we have

$$\begin{array}{l} \delta p = \kappa\dfrac{u_2^2+u_3^2+u_4^2}{u_1^2}\,\delta u_1 - 2\kappa\dfrac{u_2}{u_1}\,\delta u_2 + \\ -2\kappa\dfrac{u_3}{u_1}\,\delta u_3 - 2\kappa\dfrac{u_4}{u_1}\,\delta u_4 + 2\kappa\,\delta u_5 = 0 \end{array} \tag{39}$$

and hence with $\mathbf{n} = (0,1,0)$

$$\left[\frac{\partial \mathbf{H}^*}{\partial \mathbf{U}} + {}^t\Lambda\,\frac{\partial \mathbf{A}}{\partial \mathbf{U}}\right]\mathbf{J}_o\left\{\begin{array}{c}\delta u_1\\ \delta u_2\\ \delta u_3\\ \delta u_4\end{array}\right\} = 0 \tag{40}$$

where

$$\mathbf{J}_o = \left\{\begin{array}{cccc} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ \dfrac{u_2^2+u_3^2+u_4^2}{2\,u_1^2} & -\dfrac{u_2}{u_1} & -\dfrac{u_3}{u_1} & -\dfrac{u_4}{u_1} \end{array}\right\} \tag{41}$$

Finally from eq. 40 we obtain

$$\left[\frac{\partial \mathbf{H}^*}{\partial \mathbf{U}} + {}^t\Lambda\,\frac{\partial \mathbf{A}}{\partial \mathbf{U}}\right]\mathbf{J}_o = 0 \tag{42}$$

which is a set of four relations for Λ on the outlet boundary.

Adapting the results in literature[1] to the present case, the boundary conditions on the top and bottom walls are simply

$$n_x\lambda_2 + n_y\lambda_4 = 0 \tag{43}$$

For the kinematic adjoint, at the blade leading edge $\delta g = 0$, so that eq.29 is satisfied. At the trailing edge there is no constraint on δg and hence

$$\left[\mu\,(\mathbf{q}\cdot\mathbf{n}) + ({}^t\Lambda\,\mathbf{K})\cdot\mathbf{n}\right] = 0 \text{ on } \Gamma_b \tag{44}$$

which is the boundary condition for the adjoint equation.

Numerical solution and results

Briefly, the numerical solution of the adjoint equations is obtained by using a first-order time-dependent technique based on a finite volume formulation. The solver computes the fluxes at cell interfaces by a flux-vector splitting technique. In a similar way, the boundary conditions are imposed at the computational field edges. A more circumspect presentation of this numerical scheme is presented in a forthcoming paper.

The distributed control F^θ is null everywhere except on the blades, where it is discretized only along the radial direction. We have

$$F^\theta(x, r_i) = \mathcal{F}(r_i)\left[1 - \cos\left(2\,\pi\frac{x - x_t}{x_l - x_t}\right)\right] \tag{45}$$

so that the load on the leading ($x = x_l$) and trailing ($x = x_t$) edges is 0. For each blade considered, we have as many design parameters $\mathcal{F}(r_i)$ as the number of computational points in the radial direction.

Therefore, eq. 30 is discretized as

$$\delta\mathcal{L} = \sum_i \delta\mathcal{F}(r_i)L(r_i)(r_i - r_{i-1}) \tag{46}$$

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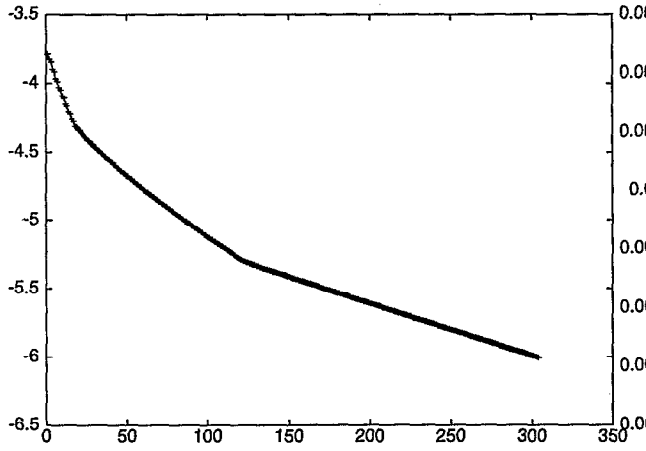


Figure 1: Gradient residuals versus optimization step.

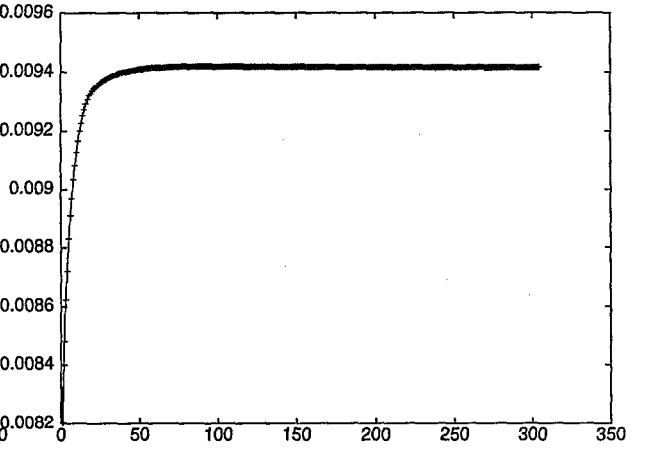


Figure 2: Thrust versus optimization step.

where

$$L(r_i) = \sum_j {}^t\Lambda(r_i, x_j) \frac{\partial \mathbf{Q}}{\partial F^\vartheta}(r_i, x_j) \left[1 - \cos\left(2\pi \frac{x_j - x_t}{x_l - x_t}\right)\right] (x_j - x_{j-1})$$

The gradient method, for example, is obtained taking $\delta\mathcal{F}(r_i) = \varrho L(r_i)$, $\varrho > 0$, so to let $\delta\mathcal{L} > 0$ in the discretization accuracy. For the applications, however, we used the conjugate gradient method.

The first application presented is a single stator with a constant radius casing. $r_t = 1$ and $r_h = 0.5$, the length of the duct is 0.5 and the grid is 40×20, so that there are 20 design variables. A flow with a constant deflection of $\epsilon = 20$ degrees is imposed at the inlet. For low Mach numbers, and with the boundary conditions imposed, the thrust maximum is expected for axial flow at the outlet. In fig. 1 we show the residuals of the increments $\delta\mathcal{F}(r_i)$ in the euclidian norm versus the number of optimization steps. During the conjugate gradient descent the parameter ϱ is divided by 2 each time the residuals decrease of an order of magnitude, thus allowing a careful search in the vicinity of the maximum. This corresponds to the slope changes seen in fig. 1. The thrust value is plotted in fig. 2. After about 50 optimization steps the thrust has reached with good approximation the maximum. We mention that the CPU time spent for the flow and the adjoint solution is remarkably reduced after the first optimization step, since the next flow and adjoint solutions are reinitialized with the last fields computed.

Given the same boundary conditions, a further numerical experiment with a reducing flow section was performed. Again we expect the maximum to be for axial outlet flow. In figs. 3 the contourlines

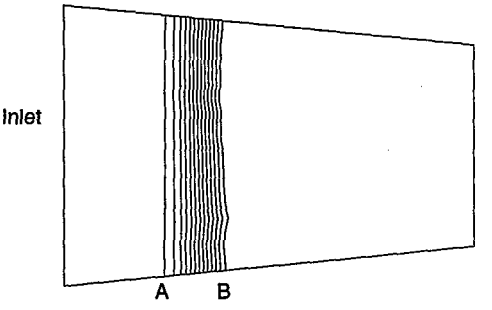


Figure 3: v component contourlines. At inlet the value is 0.19, at outlet 0.00. The blade is located between points A and B.

of the velocity component v found at the maximum are shown; it is seen that the flow is axial at outlet. The geometry of the optimized stator is that of fig. 4. To be noted the camber twist along the radius, which can not be predicted a priori.

Considered these two test cases as a preliminary validation of the method, we studied a compressor stage design. As it was mentioned in the prologue, the formulation proposed in this paper allows an easy treatment of the flow constraints, as shown by the following. In a compressor stage optimized for thrust, given the angular speed ω, it is necessary to keep constant the work done per unit volume by the rotor, that is

$$\int_{\Omega_b} F^\theta \, \omega \, r d\Omega = \text{constant} \qquad (47)$$

The increments $\delta\mathcal{F}(r_i) = \varrho L(r_i)$ are constrained to lay on the manifold determined by eq. 47. Dis-

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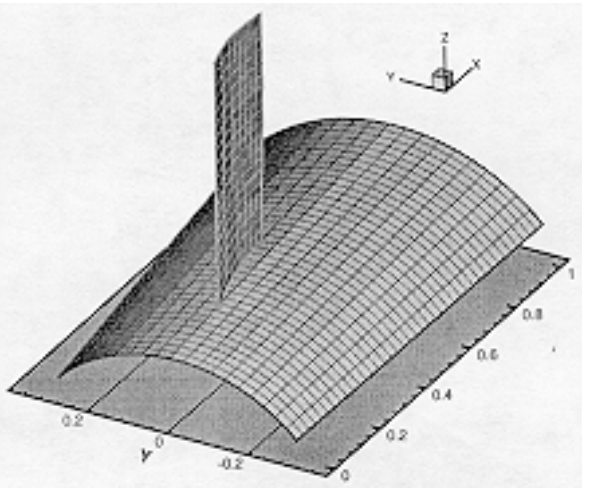


Figure 4: Blade geometry.

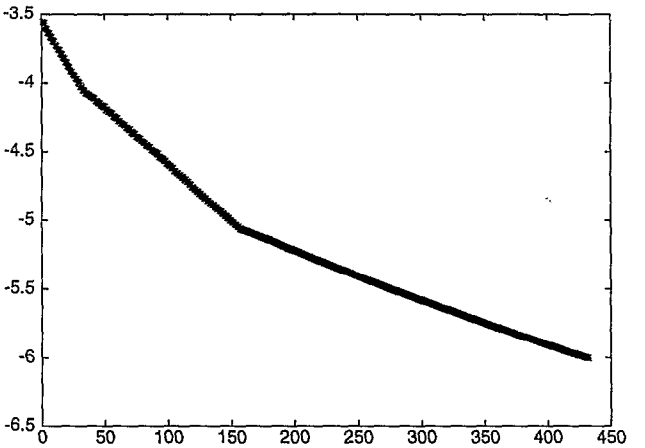


Figure 5: Gradient residual versus optimization step.

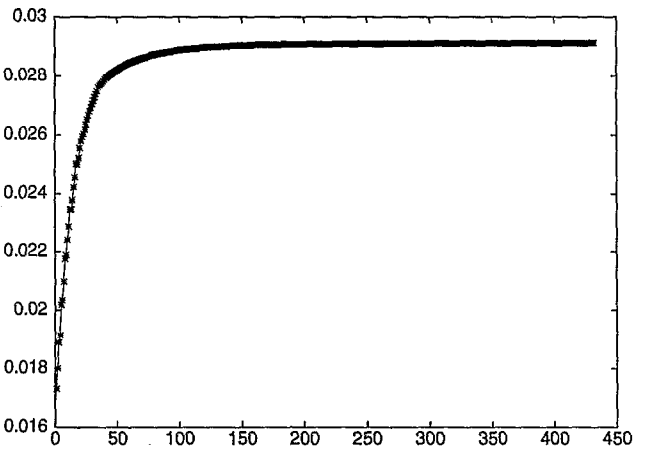


Figure 6: Thrust versus optimization step.

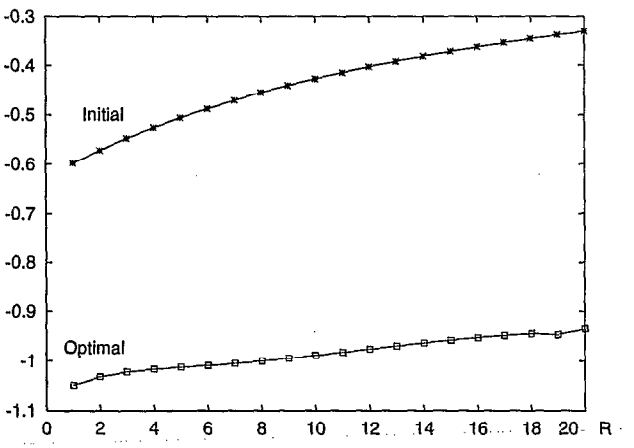


Figure 7: Forces distribution on the stator. R is the radial point considered; hub for $R = 1$. $F = \mathcal{F}(r)$.

cretizing and linearizing such equation, we obtain an hyperplane onto which the gradient can be projected in order to satisfy the constraint. This procedure is much less expensive and complicated than solving an additional partial differential equation (PDE) for the constraint as it would be necessary for usual optimization methods. Clearly the trade-off should be critically evaluated in relation to eventual geometric constraints, which in our method would be treated by an additional PDE.

The design variables for this test case, being the grid still 40×20, are 20 for the stator and 20 for the rotor; $\omega = 0.5$. The constraint on the total work done by the rotor allows very small variations of the forces distribution on the rotor itself. This is seen in the gradient components relative to the rotor that are two orders of magnitude smaller compared to those of the stator. Indeed in a different test case relative to a single rotor and not shown here, we found that for a gradient residual decreasing of two orders of magnitude, the thrust gain is very limited.

In fig. 5 and fig. 6 the gradient residuals and the thrust are plotted against the optimization step for the compressor stage test case. The gradient decreases of more than 2 orders of magnitude and the thrust increases by 70% in 50 optimization steps. As shown in figs. 7, the forces distribution on the stator changes remarkably whereas the forces distribution on the rotor do not change appreciably and are not shown.

The initial and optimal stage is in figs 8-9; the first blade from left is the stator. The angle of the flow at the entrance is $\epsilon = 20$ degrees, but in this case the optimal flow exit is not necessarily axial

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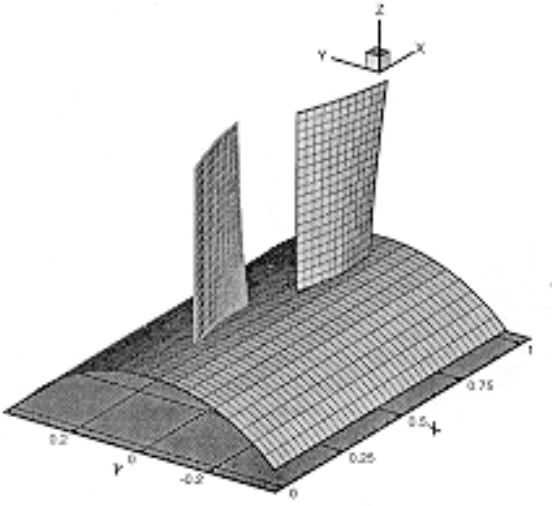


Figure 8: Compressor stage. Initial geometry.

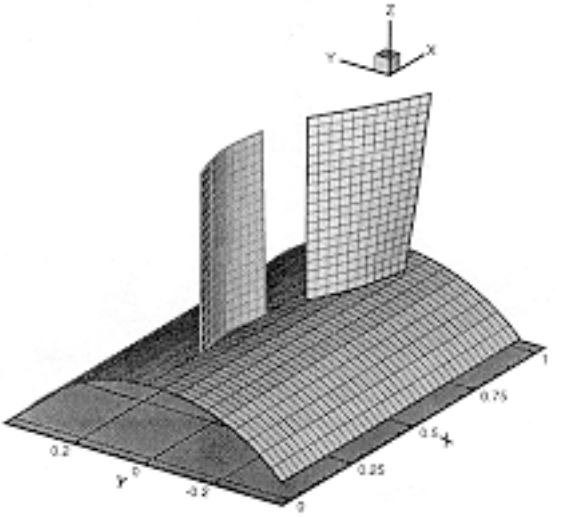


Figure 9: Compressor stage. Optimal geometry.

as there is a rotational component given to the flow by the rotor. However at the maximum the v component at the outlet is an order of magnitude smaller than that at inlet.

Note that the rotor geometry changes, but this does not correspond to a sensible forces distribution variation. The rotor simply adapts to the new flow conditions issuing from the optimal stator.

Conclusions

In this work we propose and derive an hybrid method for aerodynamic design, and apply it to turbomachinery design. It takes advantage of the inverse solution of the equations to determine optimal flows. In this sense, as opposed to shape design optimization (SDO), this methodology could be named flow design optimization (FDO). The relative advantages of using SDO or FDO must be evaluated case by case considering the number of flow constraints relative to geometric constraints. For aerodynamic components where the flow quality is vital, we advocate the preferential use of FDO.

Acknowledgments

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